Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:16 PM

Agenda

* Recap Gaussian Distribution
* Recap Central Limit Theorem
* Confidence Interval using CLT
* Confidence Interval using Bootstrapping.

** Assignments

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:12 PM

Polls

Exact number

150 ← BJP

100 ← Congress

Range : 40 – 60 seats

0 – 100 seats

Medical

CI

Q

Sample

2 / 2

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:23 PM

Quiz

Avg time taken for customers to complete is 4 min with std dev of 1 minute. Find the probability that a randomly selected customer will complete a purchase within 6 minutes ?

$\mu = 4 \text{ min}$

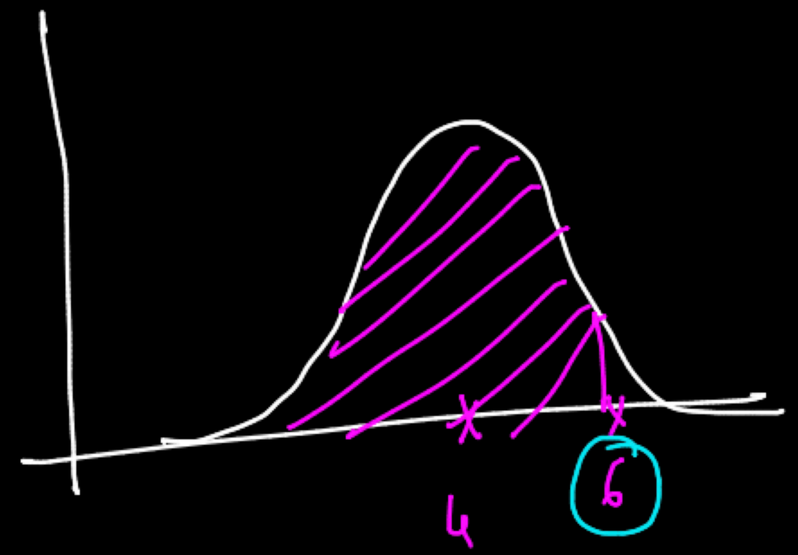
$\sigma = 1 \text{ min}$

(1) $z = \frac{6 - 4}{1} = 2$

norm.cdf(2) ↵

↑ probability

0.97



3 / 3

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:30 PM

Quiz

Avg order value on an e-commerce website is $50

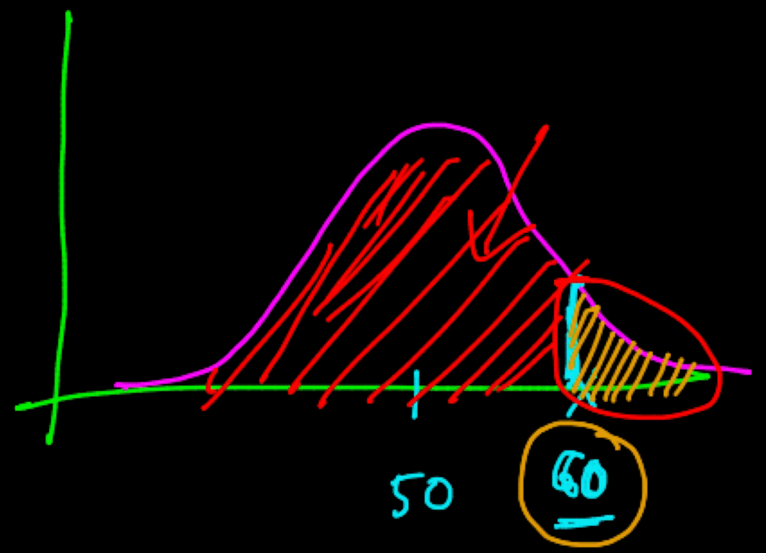
$\sigma = \$5$

cdf

probability that randomly selected order will have a value exceeding $60 ?

$\mu = 50$

$\sigma = 5$



① $z = \frac{60-50}{\sigma} = \frac{10}{5} = 2$

② $1 - \text{norm.cdf}(z)$

0.02

4 / 4

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:36 PM

Quiz

Average body temp has a mean $98.6^\circ F$

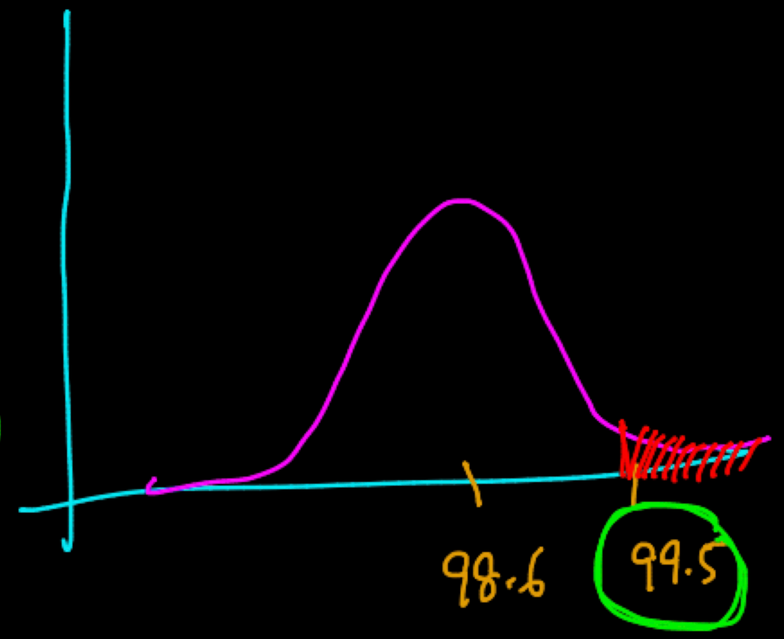
$\sigma = 0.5^\circ F$

Probability that a randomly choosen patient has a body temp higher than $99.5^\circ F$?

$\mu = 98.6^\circ F$

$\sigma = 0.5$

$z = \frac{99.5 - 98.6}{0.5} = \frac{0.9}{0.5} = \frac{9}{5} = 1.8$



1 − norm.cdf(1.8)

0.035

5 / 5

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:43 PM

Quiz

⇒ Avg time taken for customers to purchase

4 mins. σ is 1 m

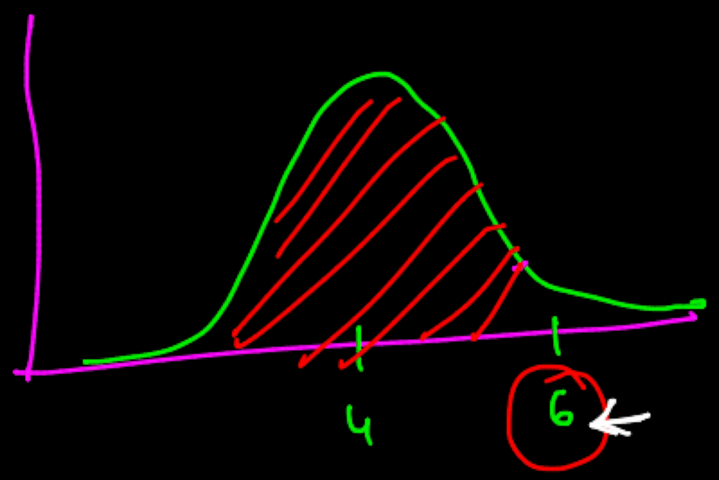
Probability that the avg time of the next 5 customers is < 6 mins?

μ = 4 min

✓ σ = 1 m

✓ n = 5

$$\text{Std_err} = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{5}}$$



$$Z = \frac{6-4}{\frac{1}{\sqrt{5}}} = ?$$

Ans

4.47

6 / 6

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:51 PM

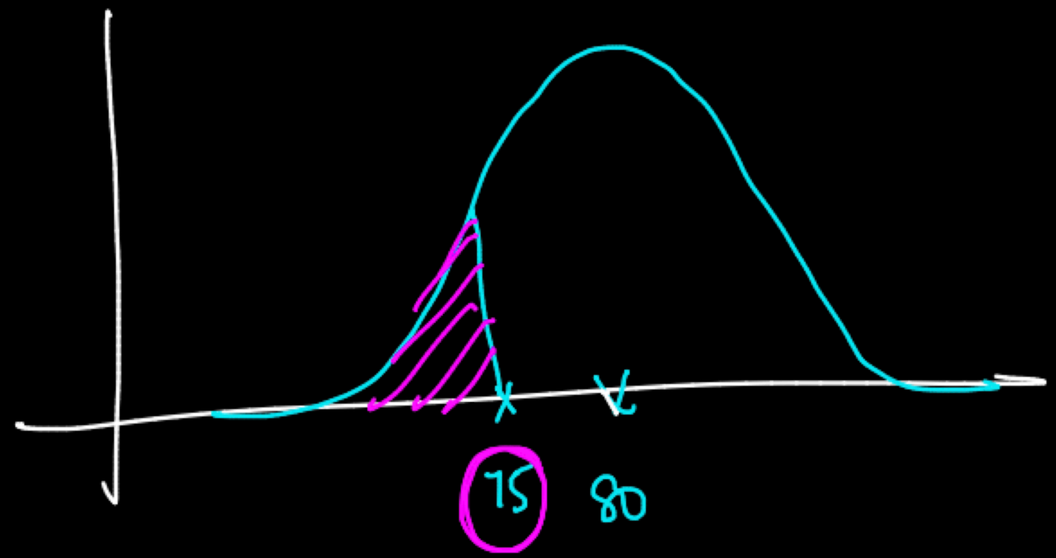
$\mu = 80$

$\sigma = 15$

$n = 50$

Probability that the avg purchase amount in the sample will be less than \$75?

$$\text{Std_err} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{50}}$$



$$z = \frac{75-80}{\frac{15}{\sqrt{50}}} = \frac{-5}{2.12}$$

$\frac{15}{\sqrt{50}} \rightarrow 2.12$

-2.357 ✓

norm.cdf(z) ↵

0.009 ✓

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 9:56 PM

Quiz

avg order value

$\mu = 50$

$\sigma = 5$

$n = 3$

Probability

avg of the next (3) orders exceeds $60

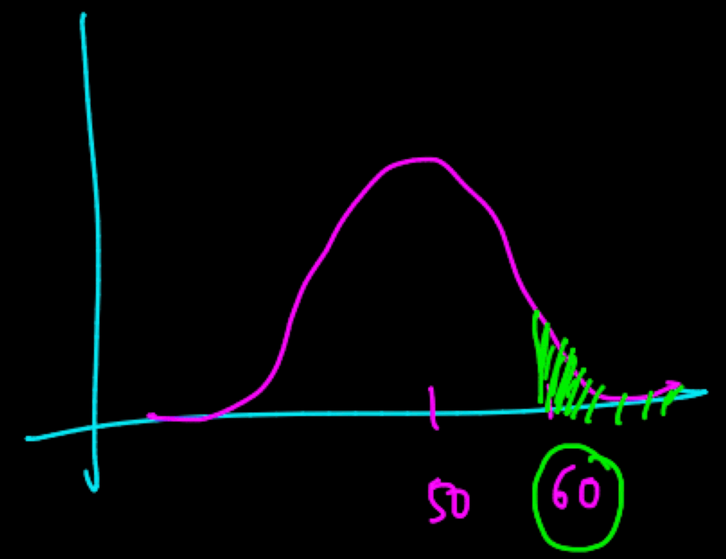
$$\text{stderr} = \frac{5}{\sqrt{3}} = 2.89$$

vs

$$z = \frac{60 - 50}{2.89} = \frac{10}{2.89} = 3.46$$

1 − norm.cdf(z)

0.00024



8 / 8

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:06 PM

Quiz

**

Weekly footy pair

mean = 1000

σ = 200

⇒ Probability that the avg weekly sales next month is more than 1110?

$\mu = 1000$

$\sigma = 200$

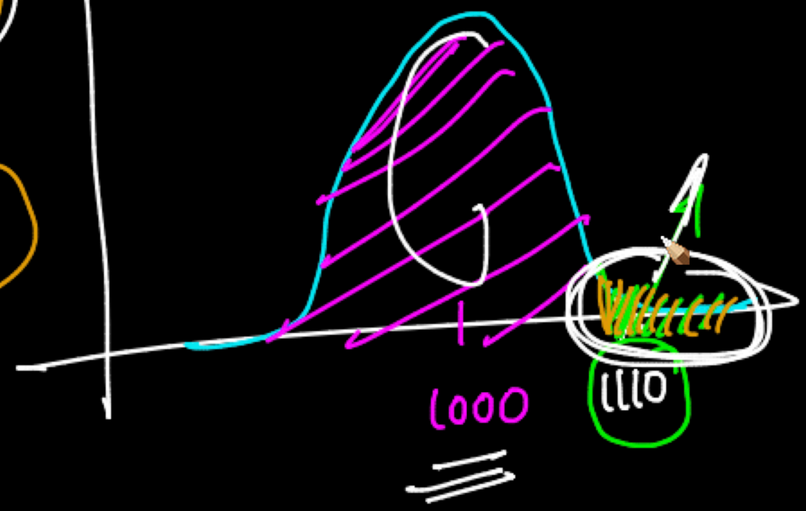
$n = 4$

std_err = $\frac{200}{\sqrt{4}} = \frac{200}{2} = 100$

$z = \frac{1110 - 1000}{100} = \frac{110}{100} = 1.1$

1 - norm.cdf(1.1)

0.1356



Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:16 PM

⇒ Confidence Interval Using CLT

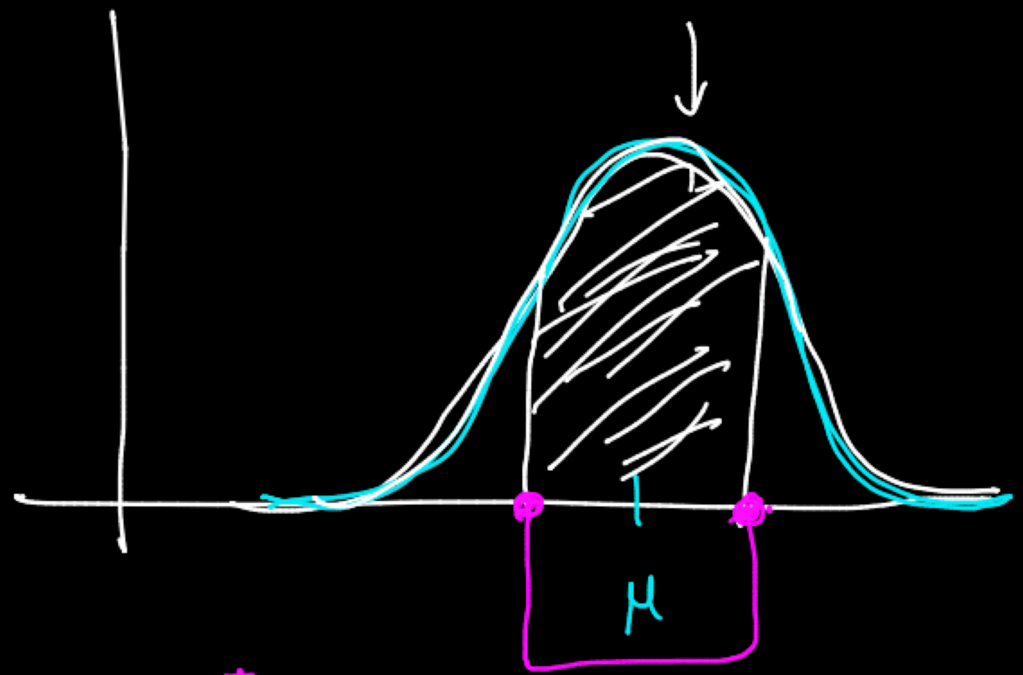
→ From some samples, you want to make some conclusion population.

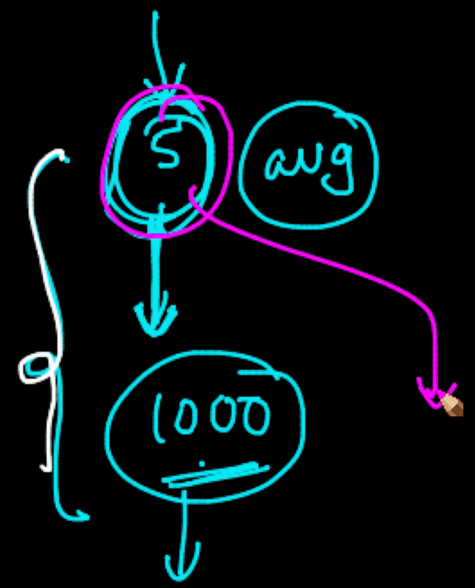
Height

The mean height of (sample of 100 people) is 65 inches, with a population std dev is 2.5 inches.

10 / 10

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:20 PM

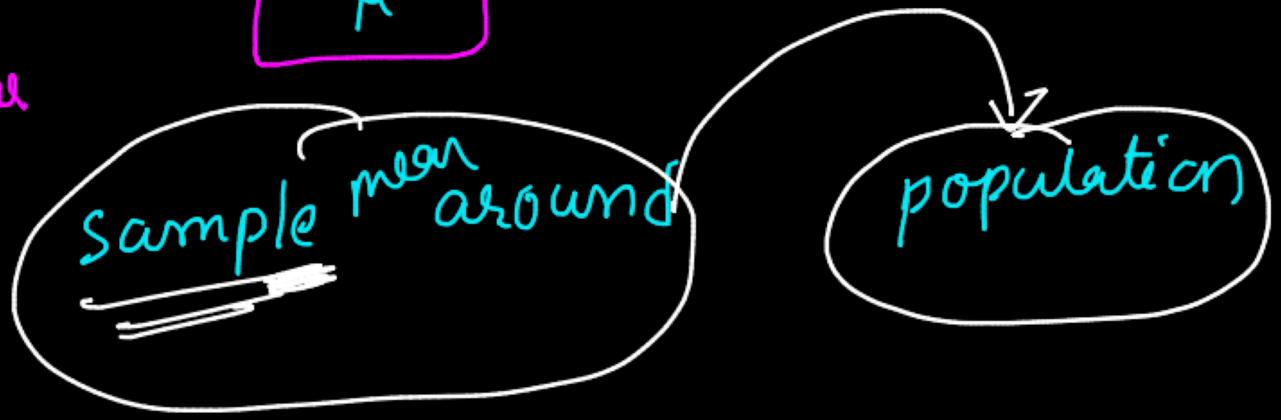




Confidence Inte

95 %

→



11 / 11

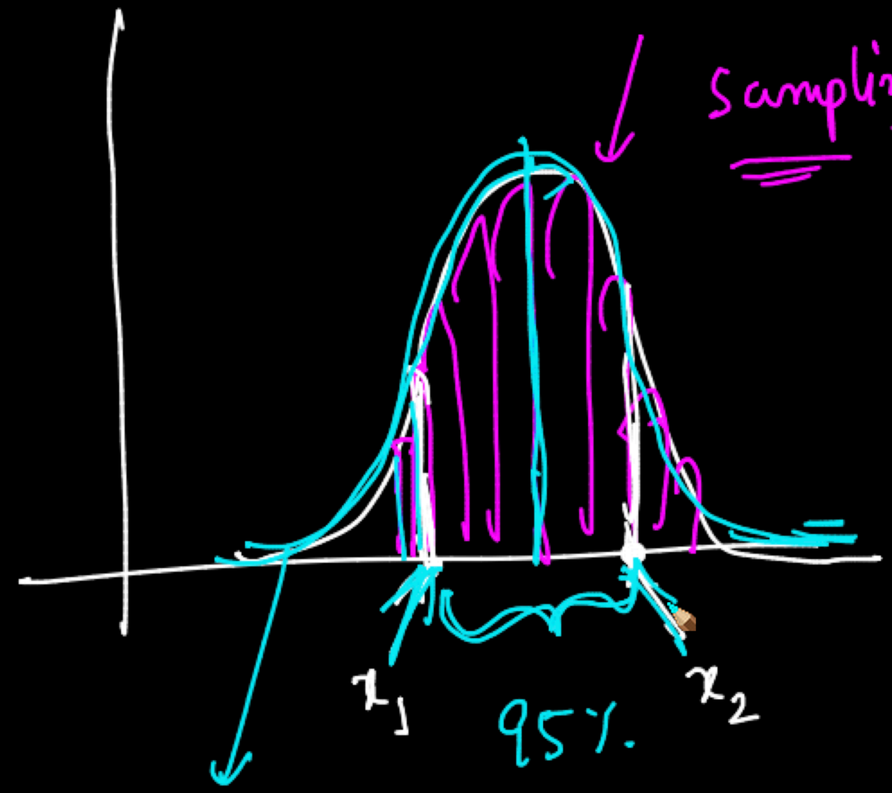
Chrome File Edit View History Bookmarks Profiles Tab Window Help
Wed 12 Jul 10:25 PM

Sample

100

65 65.5

66

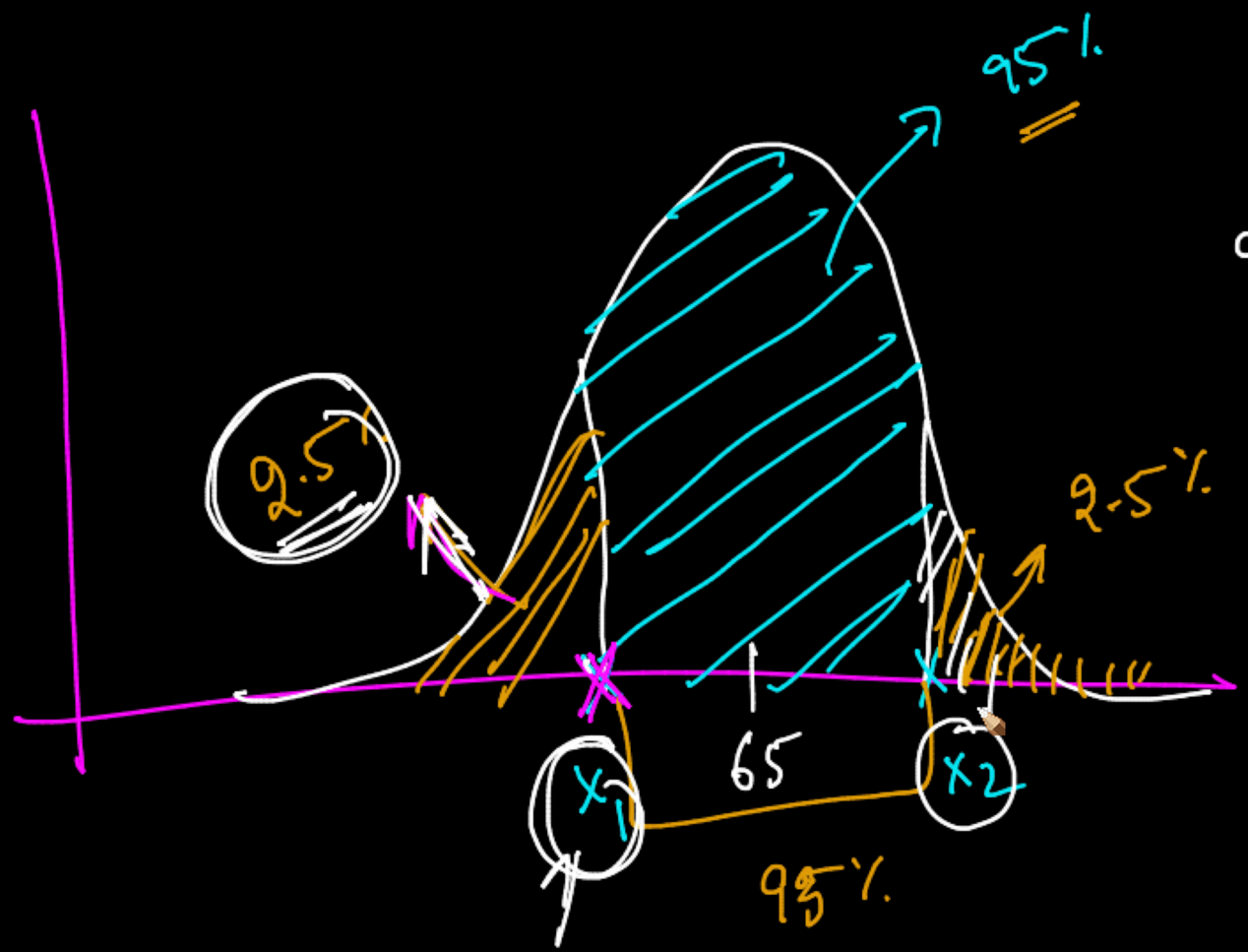


sampling

100

12 / 12

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:31 PM



65.5
64.4
63.3

9

Std = $\frac{6}{\sqrt{100}}$

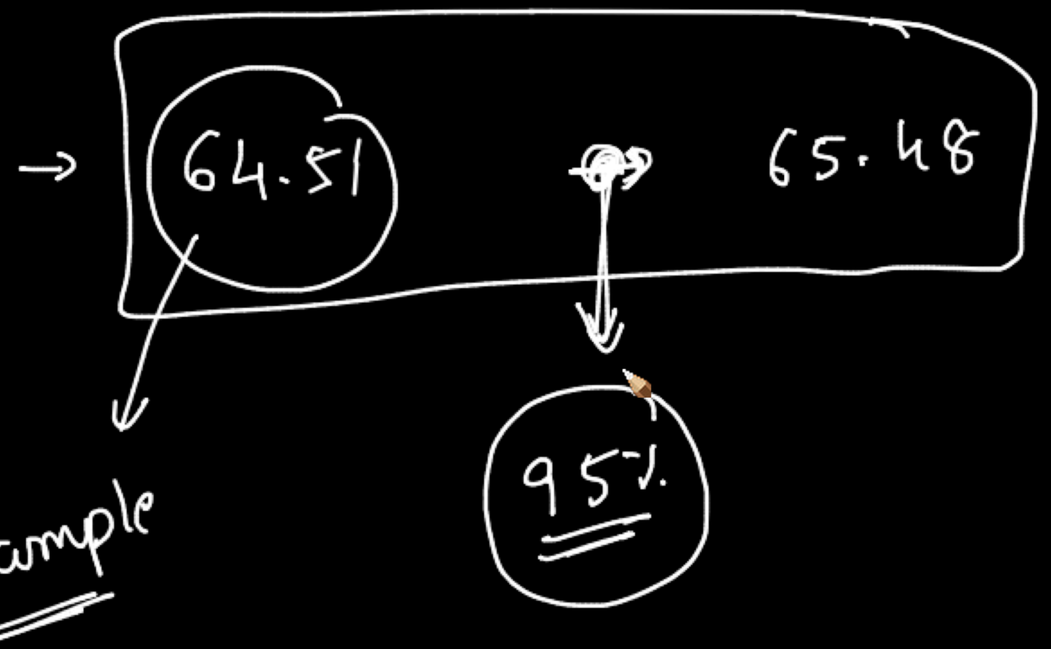
$= \frac{2.5}{\sqrt{100}}$

$= 0.25$

(z) =

z = norm. ppf (0.025)

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:34 PM



14 / 14

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:40 PM

=> Sampling

100

$\bar{x}$ = 65 inches

65

95%

2.5

2.5

$\bar{x}_1$ 65 $\bar{x}_2$

Std error $\frac{\sigma}{\sqrt{n}} = \frac{2.5}{\sqrt{100}} = 0.25$

15 / 15

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:46 PM

=> => At what interval of heights 90% CI 80%

range

95%

64.71

65.88

65.59

65.41

5%

5%

x_1 x_2

x_1 = 64.59

x_2 = 65.41

$\bar{x}$ = 65 inch

std_err = 0.25

Z_1 = norm.ppf(0.05)

= -1.644

Z_2 = norm.ppf(1-0.05)

= +1.644

16 / 16

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 10:57 PM

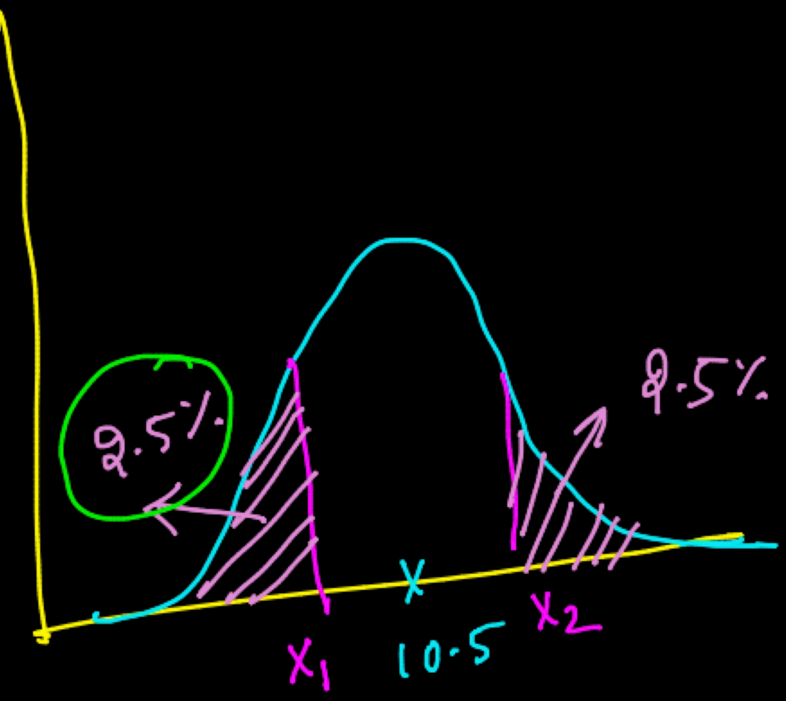
* The sample mean recovery time of 100 patients after taking a drug was seen to be 10.5 days. with Std dev of 2 days. Find the 95% CI of the true mean. population.

$n = 100$

$\bar{x} = 10.5$

$\sigma = 2$ days

$$\text{std err} = \frac{\sigma}{\sqrt{n}} = \frac{2}{10} = 0.2$$

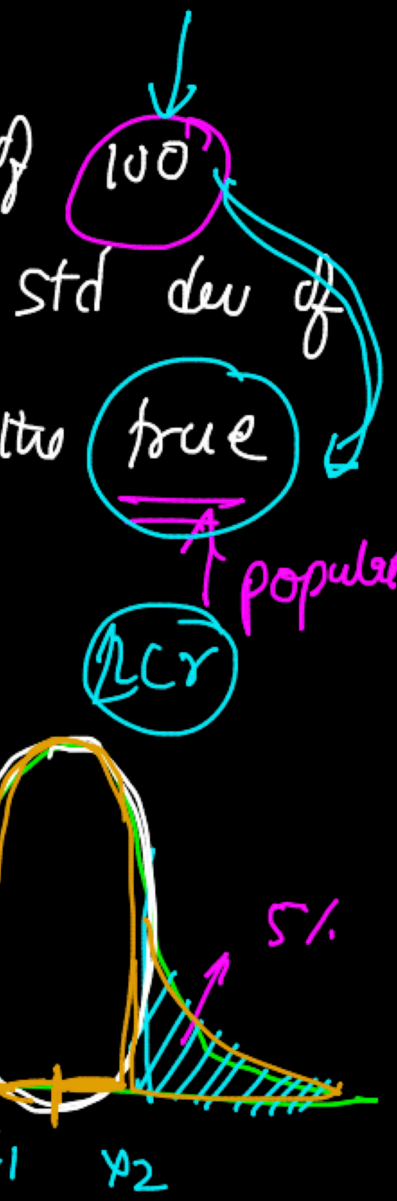


17 / 18

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 11:14 PM

Youtube watch hours

=> The mean youtube watch time of a sample of 100 students was found to be 35 hours. With a std dev of 1 hr. Construct a 90% confident interval for the true population watch time.



$n = 100$

$\bar{x} = 3.5$

$\sigma = 1h$

$z_1 =$ norm.ppf(0.05)

$z_2 =$ norm.ppf(0.95)

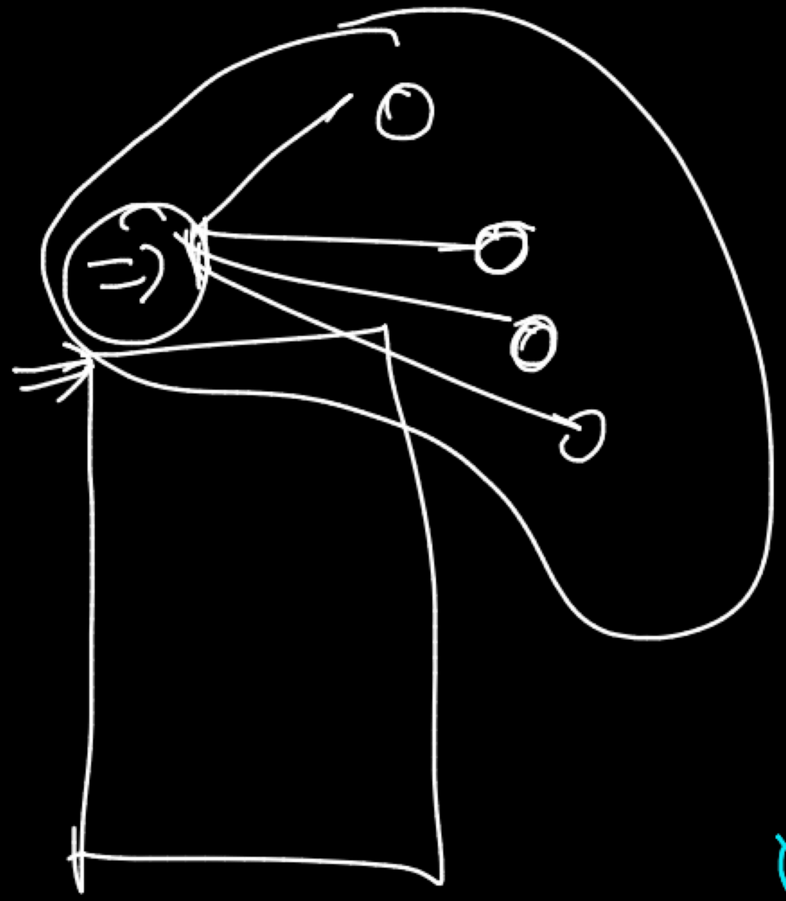
$x_1 = 3.5 + z_1 * 0.1$

$x_2 = 3.5 + z_2 * 0.1$

$3.1 \rightarrow 3.3$

$\text{Std_err} = \frac{\sigma}{\sqrt{n}} = \frac{1}{10} = 0.1$

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 12 Jul 11:37 PM



100

10000

Exit

95%

Mon

2 class

95%.

X2

9-11

Cursor selected